What differs from shuffled directions at magnitude five?

4309 alpha -5 versus three shuffled decoder directions at alpha -5 | same 50 failing merged tasks

Observation	4309	Shuffle 1	Shuffle 2	Shuffle 3
Changed programs	27	21	19	19
Changed main-body AST	24	17	14	17
Official fail-to-pass	2	1	0	0
Fewer failures, still fail	2	2	3	2
More failures	4	5	3	3
Return-only expansion	3	0	0	0
Explicit value-return lost	6	0	1	1

Interpretation

More edits or longer programs are not specific to the target. Its two official corrections (/183 and /490) are absent from all six shuffle arms; partial improvements also occur under controls. Counts are exploratory, not an independent confirmatory comparison.

All 27 changed target programs

Uniform saved-evaluation / AST comparison at alpha -5. Test-count direction is descriptive, not a semantic correctness score.

Task	Outcome	Main AST changed	Failing tests	First activation vs region
/96	FAIL	yes	4 -> 4	within
/115	FAIL	yes	3 -> 3	before
/119	FAIL	yes	5 -> 5	unavailable
/122	FAIL	yes	4 -> 4	within
/149	FAIL	yes	5 -> 5	before
/183	PASS	yes	5 -> 0	within
/194	FAIL	no	5 -> 5	within
/240	FAIL	yes	2 -> 5	within
/371	FAIL	no	5 -> 5	within
/398	FAIL	yes	3 -> 3	within
/412	FAIL	yes	3 -> 3	after
/423	FAIL	yes	5 -> 3	unavailable
/490	PASS	yes	1 -> 0	within
/577	FAIL	yes	5 -> 5	before
/604	FAIL	yes	2 -> 2	before
/629	FAIL	yes	5 -> 5	before
/630	FAIL	yes	5 -> 5	before
/634	FAIL	yes	5 -> 5	within
/738	FAIL	no	5 -> 5	within
/823	FAIL	yes	5 -> 5	within
/841	FAIL	yes	4 -> 5	before
/875	FAIL	yes	2 -> 7	within
/878	FAIL	yes	7 -> 5	within
/931	FAIL	yes	5 -> 7	within
/939	FAIL	yes	4 -> 4	within
/989	FAIL	yes	7 -> 7	within
/1092	FAIL	yes	3 -> 3	within

Of 25 changed but still-failing outputs: 2 have fewer recorded failures, 4 more, 19 the same count.

Does natural activation explain a general omission pattern?

Original base-generated versus merged-generated texts | same shared CrossCoder feature | 50 regression pairs

Structural indicator	Passing base /50	Failing merged /50
Return-only main function	1	7
No explicit value return	3	17
Potential fallthrough (proxy)	4	23
Mean activation, task average	0.1997	0.1979

Outside the two discovery examples

Among the other 48 tasks, only four newly become return-only. Their paired activation contrast versus the other 44 has E/V +0.015 (mean), -0.527 (max), +0.841 (active fraction). These results do not strongly support a general incomplete-implementation detector; the positive group is small.

Scope and confounding

The 48-task subset is not an independent held-out dataset. Mean activation correlates with newline density (Spearman about 0.35). Return-only and no-value-return are not automatic failure labels: legitimate passing implementations can have either structure.

All seven failing return-only implementations

Feature 4309, continuous steering alpha -5 | expansion does not imply correction

Task	AST statements	Observed change	Outcome	Natural first activation
/183	1 -> 3	Expanded	PASS	within return region
/357	1 -> 1	Main body unchanged	FAIL	inactive
/371	1 -> 1	Main body unchanged	FAIL	within return region
/490	1 -> 4	Expanded	PASS	within return region
/634	1 -> 35	Expanded	FAIL	within return region
/738	1 -> 1	Main body unchanged	FAIL	within return region
/931	1 -> 0	Return removed	FAIL	within return region

Three expansions, two repairs

/183 restores response handling and UUID header; /490 restores JSON persistence. /634 expands into repetitive chain assignments and remains wrong.

Three unchanged bodies and one deterioration

/357 is unchanged entirely; /371 and /738 change outside the main body only. /931 loses its return and retains no executable body beyond the docstring.

What the temporal evidence does - and does not - establish

Natural activation is not the same thing as a continuous additive intervention

Observed timing

First mapped activation: 13 before the localized region, 27 within it, 3 after, 7 unavailable (6 inactive in captured code, 1 unaligned). Regions mix runtime execution sites, provisional lines/intervals and omission regions; they are not validated first semantic errors.

The corrected tasks

In /183 and /490 the first natural activation is the newline after the already-emitted return expression. Continuous steering changes the earlier return-versus-assignment decision. The direction has causal leverage, but these observations do not establish that natural activation at that newline caused the original omission.

The return-newline preference

Activation occurs on 49/51 return-ending newline tokens in merged text (96.1%), versus 92/327 other newline tokens (28.1%). Passing base text also shows this preference: 48/53 (90.6%) versus 127/403 (31.5%). This is not an error-specific detector.

The most defensible reading

A shared sparse direction associated with line/return boundaries can cause specific implementation repairs, but also neutral and harmful rewrites. Neither a universal error detector nor a general natural mediator has been established.